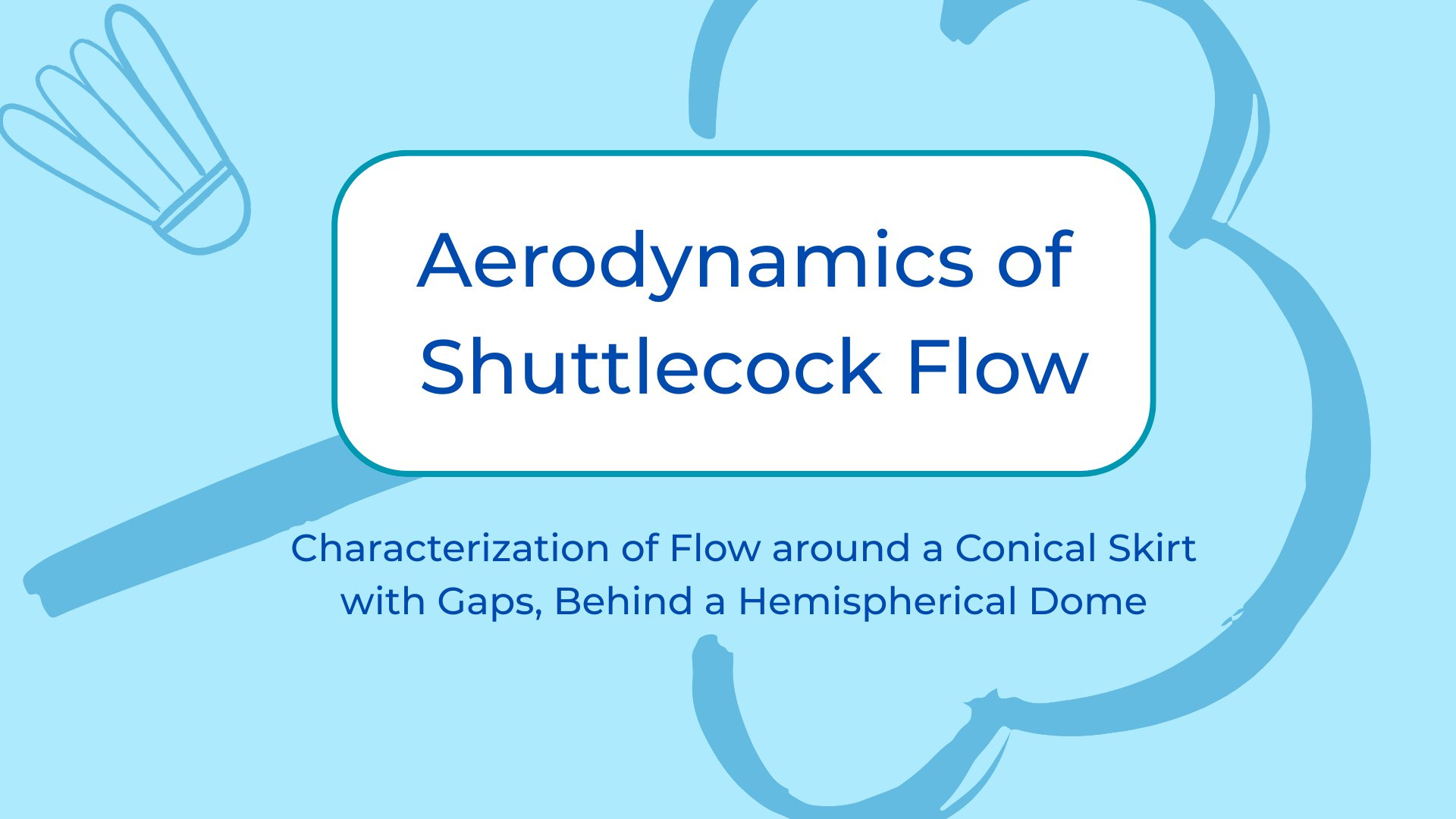




Aerodynamics of Shuttlecock Flow

Characterization of Flow around a Conical Skirt
with Gaps, Behind a Hemispherical Dome

WHAT IS AERODYNAMICS ?

Aerodynamics is a branch of fluid mechanics that focuses on the behavior of air (or other gases) in motion and the forces that arise when air interacts with solid objects. This field is essential in understanding how objects travel through the atmosphere and how air flows around them.

At its core, aerodynamics examines four fundamental forces:

- **Lift** – The upward force that opposes gravity (crucial in flight).
- **Drag** – The resistive force that opposes motion through air.
- **Thrust** – The forward force that propels an object.
- **Gravity (Weight)** - The downward force pulling objects toward the Earth.

Aerodynamics isn't just for aircraft it's critical in sports, automotive engineering, architecture, and even medicine.

- For example:
- In **Formula 1**, aerodynamics can win or lose races.
- In **cycling**, reducing air resistance (drag) can cut race times.
- In **badminton**, it governs how the shuttlecock behaves after a hit.

Why Study Shuttlecock Aerodynamics?

1. To Understand and Improve Flight Behaviour
2. To Design Better Shuttlecocks
3. To Improve Performance in Competitive Play
 - Small aerodynamic differences can affect:
 - Accuracy and timing of shots
 - Consistency across different brands or models
4. Practical Applications Beyond Badminton
 - Insights gained from studying shuttlecock aerodynamics can be applied to:
 - Parachute design
 - Drones and micro-air vehicles
 - Air braking systems
 - Architectural ventilation

Drag Force – The Key to Shuttlecock Flight

What It Is:

- Drag is the resistance force caused by air pushing against the shuttlecock as it moves.
- It slows the shuttlecock after a smash or serve.

Variables:

- F_D : Drag force
- C_d : Drag coefficient (depends on shape)
- ρ : Air density
- A : Cross-sectional area
- V : Speed of shuttlecock

$$F_D = \frac{1}{2} C_d \rho A V^2$$

Why It's Used:

- High drag slows the shuttle quickly.
- Helps it flip cork-first and remain stable.
- Ensures safe and controlled flight.



Fun fact

The shuttlecock is the **slowest and fastest object** in racket sports!

It can leave the racket at over 400 km/h — but slows down incredibly fast because of air resistance



Wake and Flow Separation – The Invisible Effect

What it is

The shuttlecock skirt shape causes the air to separate from the surface, forming a wake (turbulent region behind it). This creates form drag, a major source of resistance.

Mathematically:

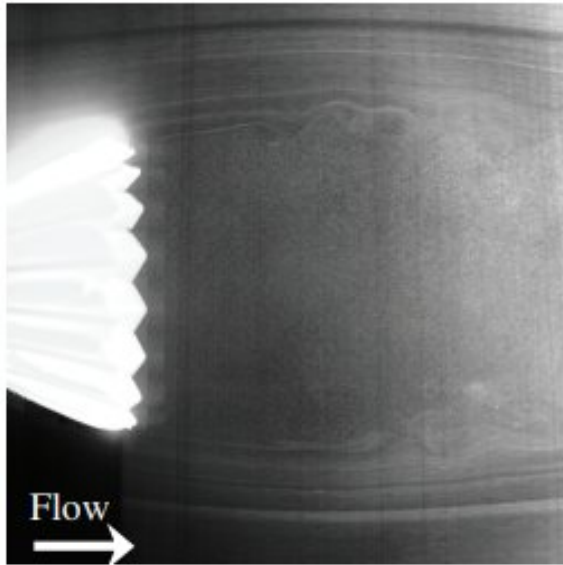
- Pressure drag arises due to difference in pressure between the front and back:

$$\Delta P = P_{\text{front}} - P_{\text{rear}} \longrightarrow F_D = \Delta P \cdot A$$

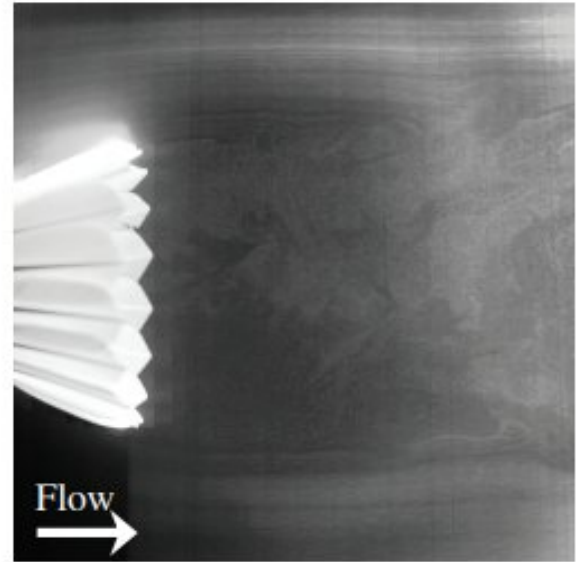
- This is especially important in bluff bodies like shuttlecocks

We can model how wide the wake is and how fast vortices are shed using the

Strouhal Number (St)



(a)



(b)

Fig.6 Flow visualization around the shuttlecock in X-Z plan for $Re = 8.6 \times 10^3$ ($\alpha = 0^\circ$) for (a) shuttlecock without gap; (b) ordinary shuttlecock.

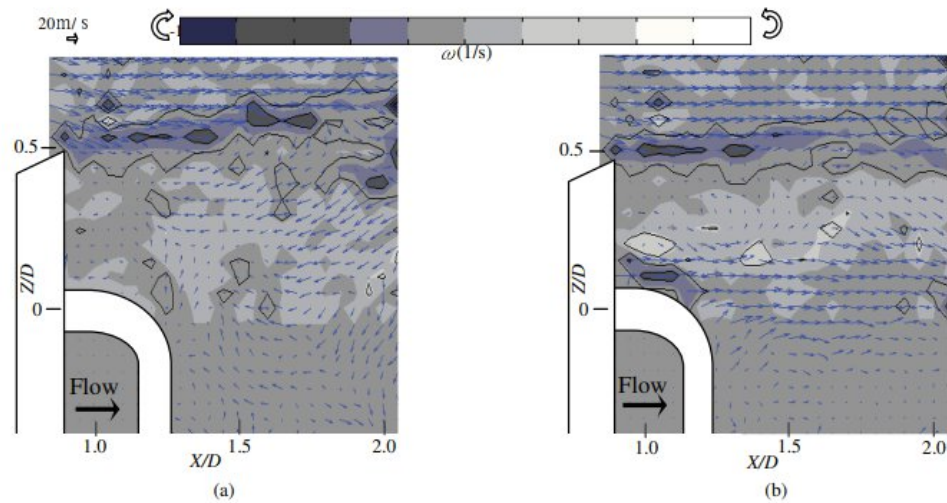
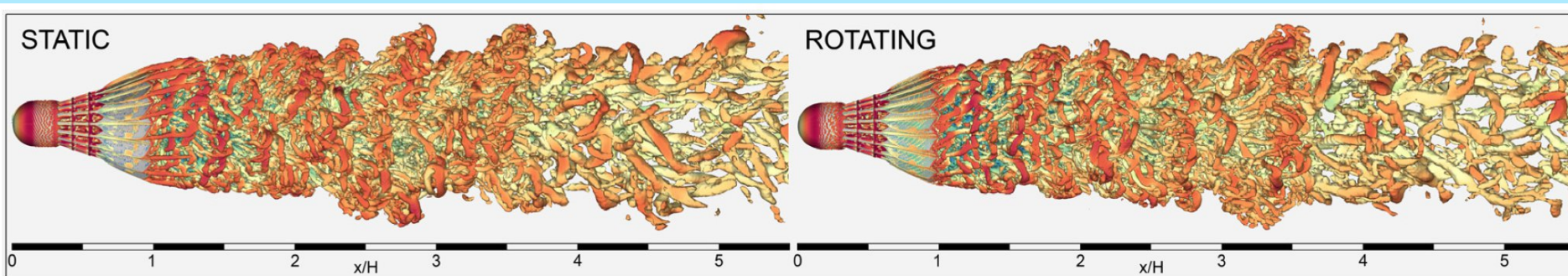


Fig.7 Density map of vorticity in X - Z plan for $Re=8.6 \times 10^4$ ($\alpha=0^\circ$) for (a) shuttlecock without gap; (b) ordinary shuttlecock.





Strouhal Number (St):

A dimensionless number used to describe how frequently vortices are shed behind the shuttlecock.

Where:

f: vortex shedding frequency

L: characteristic length (e.g., skirt diameter)

$$St = \frac{fL}{V}$$

- In sports (like badminton), engineers use the Strouhal number to control wake behavior.
- A consistent shedding frequency helps stabilize objects in flight (like keeping the shuttlecock cork-first).



Lift force

What is it ?

Lift is the aerodynamic force that acts perpendicular to the direction of airflow. It's what helps shuttlecocks stay aloft against gravity

How it works in a shuttlecock

- The conical skirt creates a pressure difference between and bottom surfaces.
- Air flows faster under the skirt, causing lower pressure below.
- This difference generates lift, slowing the fall and keeping shuttlecock stable

$$L = \frac{1}{2} \rho V^2 A C_L$$

the top

Why it's important:

- Lift ensures the shuttlecock doesn't just drop like a rock.



Terminal Velocity

The shuttlecock quickly reaches a maximum falling speed due to drag balancing weight.

Where:

- V_t = terminal velocity
- m = mass of the shuttlecock
- A = cross sectional area
- g = acceleration due to gravity
- C_D = drag coefficient
- ρ = air density

$$V_t = \sqrt{\frac{2mg}{C_D\rho A}}$$

Use:

Helps understand how fast a shuttle can fall from a high lob or drop shot.



Fun fact

Shuttlecocks Reach Terminal Velocity in Just 1 Second

- After a smash, air drag slows the shuttlecock so fast that it reaches terminal velocity within about a second.
 - This is why shuttle shots drop suddenly and steeply!



Reynolds Number – Understanding Airflow

What It Is:

Determines whether air clings to surface (laminar) or separates (turbulent).

Variables:

- ρ : Air density
- V : Velocity
- D : Characteristic length (e.g., skirt diameter)
- μ : Dynamic viscosity

$$Re = \frac{\rho V D}{\mu}$$

Why It's Used:

- Helps predict if the airflow will separate and form a wake
- Determines how much turbulence surrounds the shuttle
- Designers simulate this to predict drag and flip stability

Pressure Effects in Feather, Plastic, and Gapless Shuttlecocks

Feather Shuttlecock

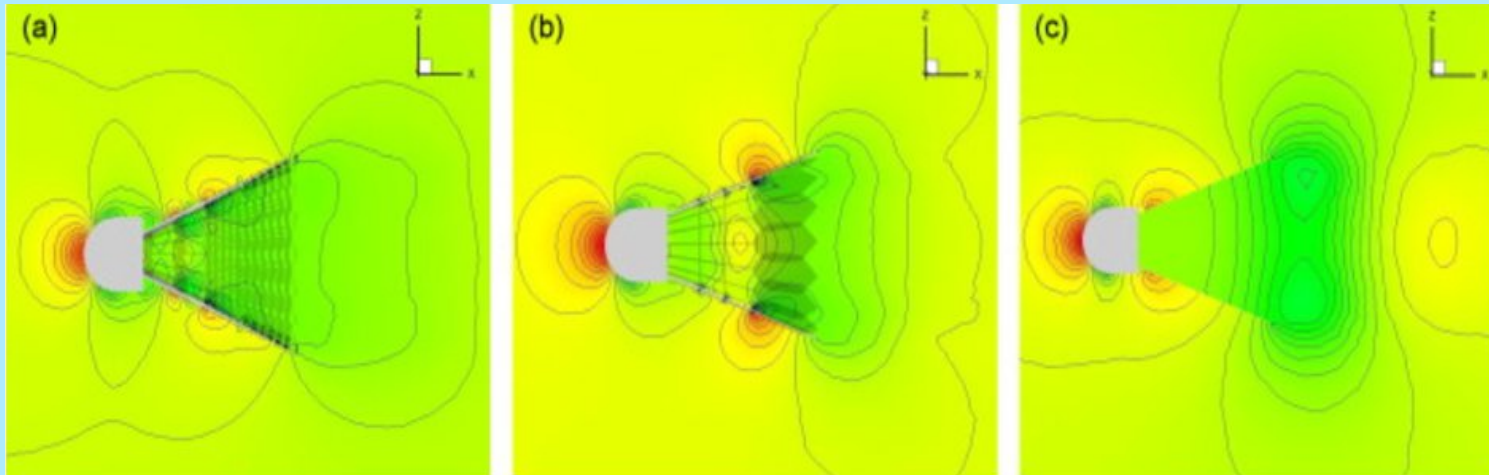
- The feather shuttlecock's complex flow dynamics can be analyzed through potential flow theory in the complex plane.
- The gaps between feathers create periodic discontinuities that **act like singularities** in the flow field, disrupting the otherwise smooth potential flow solution. These gaps **induce vortex shedding** that manifests as branch cuts in the complex potential function $F(z) = \phi + i\psi$.
- The pressure coefficient **C_p shows strong variations** due to this discontinuous flow separation, with particularly low pressure regions forming behind each feather gap.
- This aerodynamic behavior explains why feather shuttles maintain **more stable flight trajectories** despite their higher drag characteristics compared to synthetic alternatives.

Plastic Shuttlecock

- The plastic shuttlecock's synthetic vanes present a more **regular geometric pattern** that can be modeled using conformal mapping techniques.
- The pressure coefficient distribution shows **less extreme variations** than in feather shuttles, as the flow separation occurs more predictably at each vane edge.
- This regularity in the complex flow field results in moderately negative C_p values on the leeward side, producing less drag but also slightly **less rotational stability** during flight.
- The mathematical treatment reveals why plastic shuttles fly faster but with less precision than feather ones.

Gapless Cork

- A gapless shuttlecock would represent a classic potential flow problem solvable through conformal mapping of a hemispherical body.
- Without flow separation points, the complex potential $F(z)$ would remain **analytic** throughout the domain, **producing a smooth, symmetrical pressure coefficient distribution**.
- The C_p would follow ideal potential flow predictions, with maximum pressure at the stagnation point and gradually decreasing values over the hemispherical surface.
- While this configuration would theoretically minimize drag, the absence of vorticity-generating gaps **eliminates the stabilization** crucial for proper shuttlecock flight behavior.
- This explains why real shuttlecocks must incorporate gaps despite their negative impact on aerodynamic efficiency.



Hence Feather shuttles have big pressure difference that slow them a bit but make them spin and fly straight, making them the best choice.

Whereas plastic and solid shuttles have smaller pressure changes, making them wobble more.



Fun fact

The shuttlecock's unique flip phenomenon due to vortex shedding makes it the only sports projectile that actively corrects its own flight instability through aerodynamics

Importance of Aerodynamics in Shuttlecock Performance and Design

1. Performance: Flight Control & Stability

- The shuttlecock must remain stable and fly dome-first no matter how it's hit.
- Aerodynamics ensures:
 - Rapid deceleration (due to high drag), preventing it from overshooting.
 - Consistent trajectory, so players can predict and respond accurately.
 - Resistance to wobbling, thanks to wake formation behind the conical skirt.

2. Design: Shape, Material, and Structure Optimization

- Designers rely on aerodynamic studies to fine-tune:
 - Skirt angle and shape (affects drag and stability)
 - Gap size between feathers/plastic strips (modifies wake and airflow)
 - Weight distribution (affects lift/drag balance and spin)
 - Material choice (feather vs. synthetic affects surface roughness and air interaction)

Fun fact

A feathered shuttlecock uses exactly 16 feathers!

- Only feathers from the left wing of a goose or duck are used , so they curve the right way!

EVOLUTION OF THE SHUTTLECOCK

- has an open conical shape formed by feathers or plastic (or a synthetic alternative) embedded into a rounded cork (or rubber) base.
- Regardless of initial orientation, it will turn to fly cork first, and remain in the cork-first orientation.
- most modern and widely used form of shuttlecock until recently was formed from 16 or so overlapping feathers, usually goose or duck, embedded into a rounded cork base.
- For a long time, geese feathers were used for international play and duck feathers for club play.
- The cork used to be covered with thin leather and feathers from right or left wings only were used in each shuttlecock to ensure optimal flight.
- Owing to the ethical concerns, in 2021, natural feather shuttlecocks were replaced by synthetic ones in all the tournaments sanctioned by the Badminton World Federation.

For years the shuttlecock has gone through various structural and materialistic changes.



Circa 1840, Made In India - 35 feathers wrapped in a velvet base



Circa 1890, Made in France- 20 feathers into a leather base (no stitching)



Circa 1915, Made by F.H Ayres, London .Stitching under the plume of the feather as well as the base of the stem



Circa 1935, experimental shuttlecock using teased natural gut but failed as the flight was erratic.

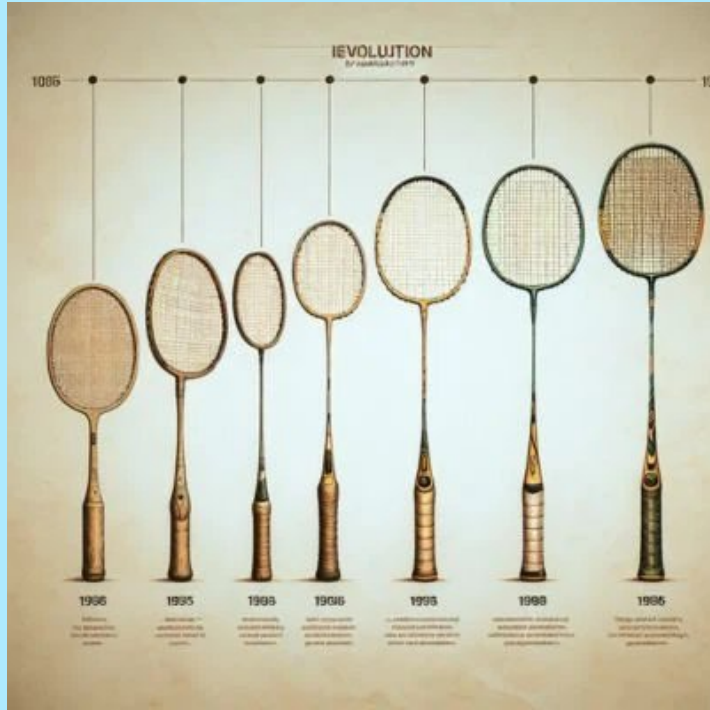


Circa 1963, Made by RSL- Feathers were punched in a different configuration to give a more stable flight.



Circa 1993, modern goose feathered shuttle. Fixed into a two-piece cork base.

EVOLUTION OF THE RACQUET



- The Earliest were made entirely of wood, making them heavy and unwieldy.
- By the mid-20th century, the introduction of Aluminum Frames offered lighter and more durable alternatives.
- Introduction of Carbon-fiber composites in late 20th century made racquets incredibly lightweight, stiff, and responsive
- Modern Racquets now feature aerodynamic designs, advanced string technologies, and ergonomic grips, catering to the demands of today's fast-paced game.

ØKey Fluid Dynamic Concepts

Concept	Simple Meaning	How it happens in Shuttlecock
Flow Separation	Air cannot stick to surface, moves away	Air leaves surface at dome and skirt
Wake	Disturbed, swirling air behind	Big turbulence behind the shuttle
Pressure Drag	Slowing force from air pressure	High pressure in front, low pressure behind
Porous Flow	Air passes through holes/gaps	Air goes through the gaps in the skirt
Passive Stability	Natural alignment in air	Skirt keeps shuttle pointing nose-first
Reynolds Number Effect	How smooth/turbulent air is	Air starts smooth, becomes turbulent behind

1. Mass Conservation and the Continuity Equation

- Mass conservation asserts that the total mass of a fluid remains constant within a control volume, assuming no sources or sinks. For incompressible flows (e.g., air at subsonic speeds), this principle simplifies to the continuity equation:

$$\nabla \cdot \mathbf{u} = 0$$

- Here, $\mathbf{u}$ represents the velocity vector field. This equation states that the divergence of velocity-the net rate of outflow per unit volume-is zero, ensuring that the mass flow rate entering a region equals the mass exiting it
- The principle of mass conservation ensures that the mass flow rate of air entering a control volume equals the mass flow rate exiting it.
- In the case of a shuttlecock, air bleeds through the gaps in the conical skirt.
- This bleed modifies the airflow pattern and wake structure behind the shuttlecock.
- The altered wake directly influences the drag characteristics and flight stability.

2. Momentum Conservation and the Navier-Stokes Equations

- The Navier-Stokes equations describe momentum conservation for viscous fluids, accounting for forces like pressure gradients, viscous stresses, and inertial effects. For incompressible flow, they simplify to:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

- ρ : Fluid density
- p : Pressure
- μ : Dynamic viscosity
- $\mathbf{f}$: External forces (e.g., gravity)

- These equations govern velocity and pressure fields, capturing phenomena like turbulence, flow separation, and vortex shedding.
- Key aerodynamic effects such as flow separation, vortex formation, and wake development are governed by these equations.
- Gaps in the skirt generate air jets, influencing drag and pressure distribution.
- CFD (Computational Fluid Dynamics) simulations use these equations to analyze the shuttlecock's aerodynamic behavior.

ØSummary Table with Key Equations and Examples

Concept	Equation/Formula	Example/Application
Mass Conservation	$\nabla \cdot \mathbf{u} = 0$	Continuity of airflow through gaps
Navier-Stokes	$\rho(\partial \mathbf{u} / \partial t + \mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu \nabla^2 \mathbf{u}$	CFD simulation of flow around shuttlecock
Reynolds Number	$Re = \frac{\rho U L}{\mu}$	$Re \approx 2 \times 10^5$ at 50 m/s shuttlecock speed
Drag Force	$F_D = \frac{1}{2} C_D \rho U^2 A$	Drag increases with gap size, affects shuttlecock stability
Terminal Velocity	$V_T = \sqrt{\frac{2mg}{C_D \rho A}}$	Used to model shuttlecock flight trajectory
Trajectory Equations	$m \frac{d^2 x}{dt^2} = -D \cos \theta + L \sin \theta$	Predict shuttlecock flight path with initial velocity and angle
	$m \frac{d^2 y}{dt^2} = -D \sin \theta - L \cos \theta - mg$	

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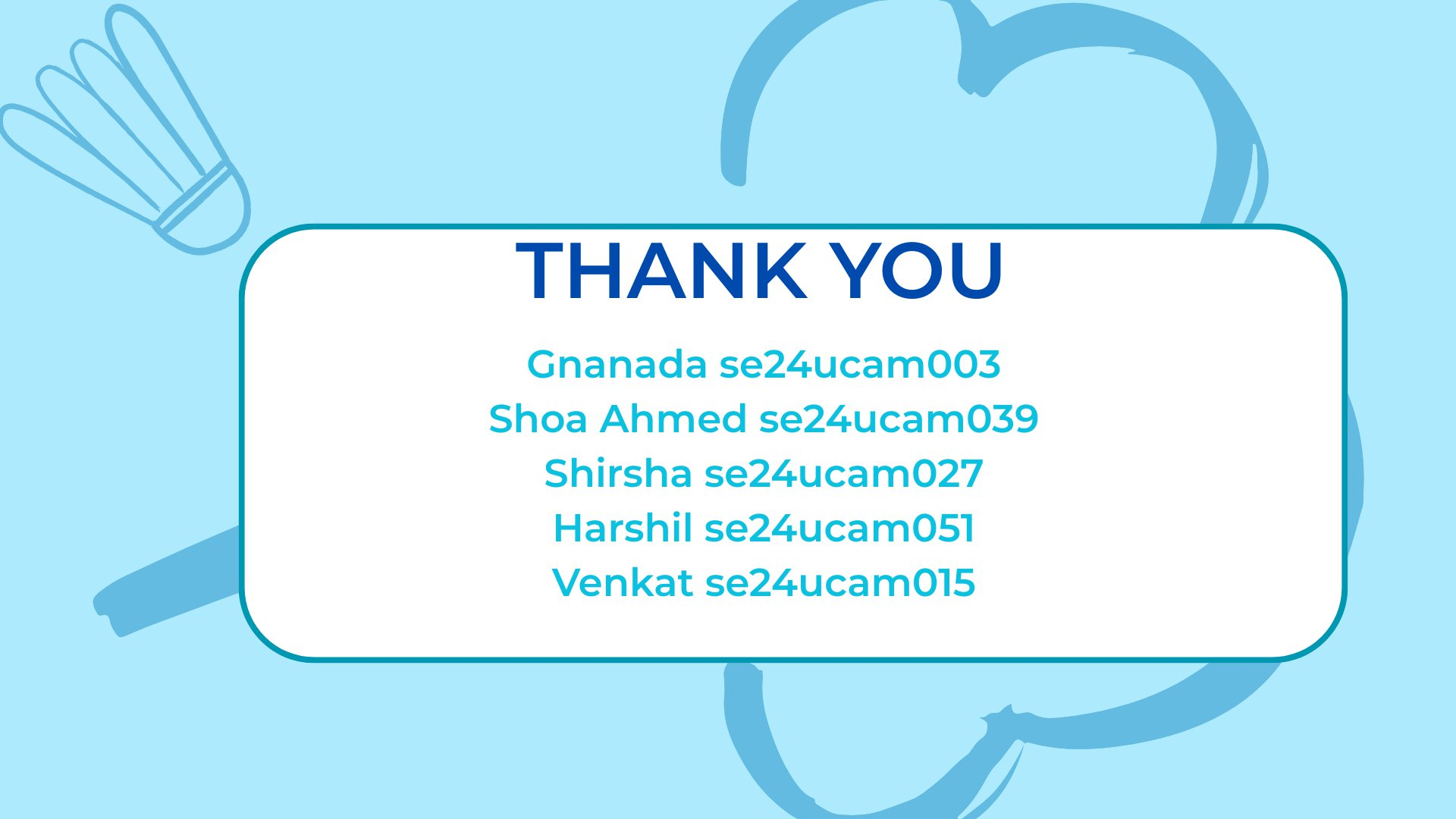
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